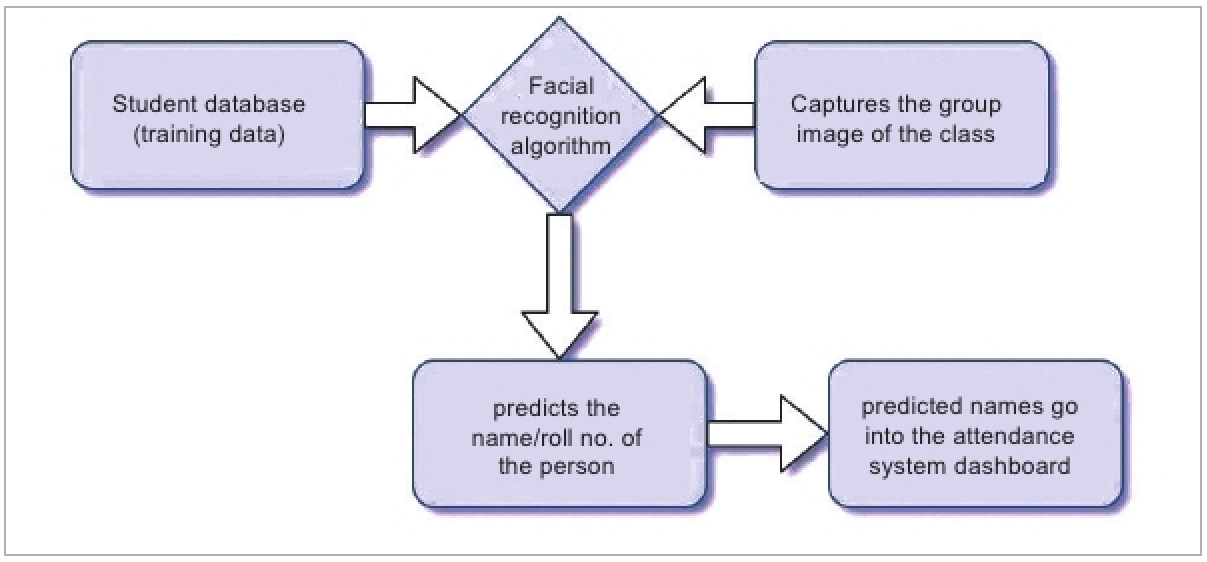

Figure 1: Diagrammatic representation for the first approach

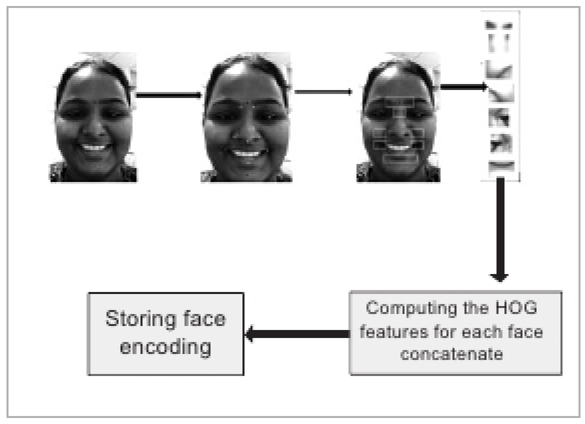

Figure 2: Diagrammatic representation for the second approach

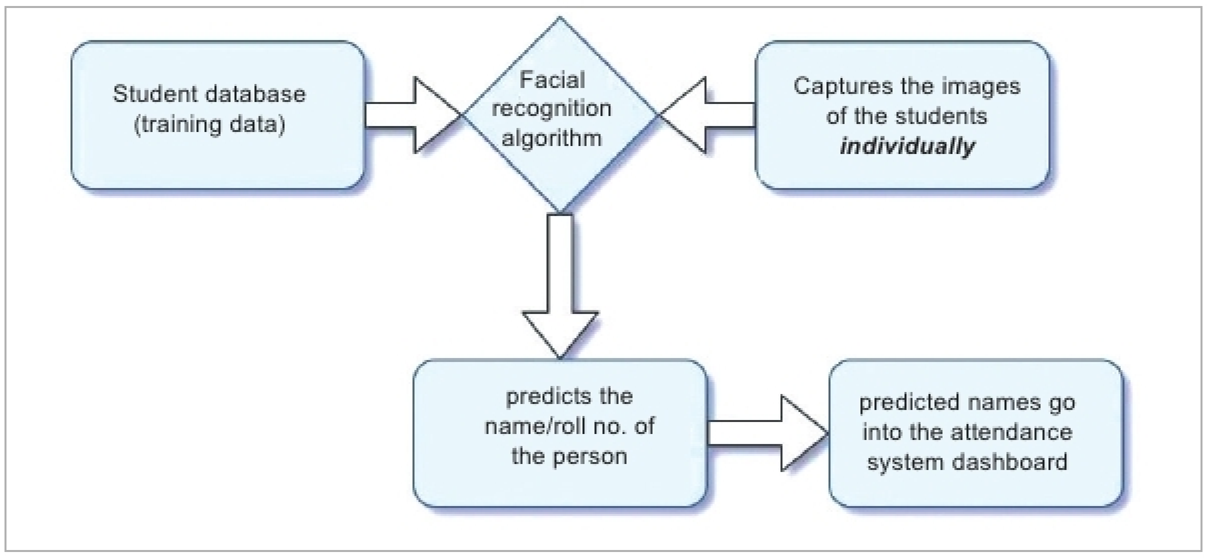

Figure 3: How HOG works

```
path = "/home/srujana/Downloads/Photos/"
dirs = os.listdir( path )

# This would print all the files and directories
known_face_encodings = []
known_face_names = []
sep = '.'
for file in dirs:
    print file
    image1 = face_recognition.load_image_file(path+file)
    image1_face_encoding = face_recognition.face_encodings(image1)[0]
    known_face_encodings.append(image1_face_encoding)
    name = file[:-4]
    known_face_names.append(name)
```

Figure 4: Code snippet for extracting the face encodings

```
        ret, frame = video_capture.read()


        small_frame = cv2.resize(frame, (0, 0), fx=0.25, fy=0.25)


        rgb_small_frame = small_frame[:, :, ::-1]


        if process_this_frame:

        face_locations = face_recognition.face_locations(rgb_small_frame)
        face_encodings = face_recognition.face_encodings(rgb_small_frame, face_locations)
```

Figure 5: Code snippet for real time capture

and increases success rates. With the advent of face recognition, numerous techniques have been incorporated in attendance management systems.

The various technologies used to take attendance

Students' attendance using RFID tags: RFID stands for 'radio frequency and identification'. RFID methods use radio waves to accomplish this. At a simple level, RFID systems consist of three components — an RFID tag or smart label, an RFID reader and an antenna. The job of the reader is to convert radio waves to some usable form of data. Information that is collected from these tags is transferred through a communications interface to a host computer system, where the data can be stored in a database and analysed later. The RFID device serves the same purpose as a bar code or a magnetic strip on the back of a credit card or ATM card. Just as a barcode needs to be scanned to get the information, similarly, the RFID device must be scanned to retrieve the data. Each student is required to have her/his student RFID card with them when entering the classroom. The back-end data is managed by custom software.

RFID has a lot of advantages – it can store more information as compared to bar codes and it can provide locations to the reader along with the ID. These tags can be both read-only and read/write (so the tags can be reused). RFID's several disadvantages include that it requires a battery, privacy is